

CHAPTER 7

Postmortem Examination of the Asian Elephant

Protocol, Equipment, Systematic Examination & Differential Diagnosis

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Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Wildlife Health Management — Context

Wildlife health is an important but still-developing discipline that is often overlooked, even though it is a key component of wildlife conservation.

Definition

Definition (Shrivastav, 2010): Wildlife Health Management is a branch of veterinary science dealing with veterinary interventions in wild fauna — health monitoring, disease management, scientific immobilization, and habitat manipulation — to achieve conservation strategies that maintain a healthy population or conserve a particular species.

2. Introduction to Postmortem Examination

Terminology

Term	Definition
Necropsy	Postmortem examination of animals and birds.
Autopsy	Postmortem examination of humans.

Key Principles

- PM examination must be conducted as per a **standard protocol** and with **adequate facilities**.
- The carcass of a wild animal is **seldom found in a condition suitable** for PM examination — early detection and prompt response are critical.
- **Who should perform it?** An experienced wildlife veterinarian, or a veterinary pathologist with exposure to wildlife.
- Postmortem examination is an **art of disease diagnosis** — a scientific, systematic requirement for recording gross pathological changes in the carcass.
- **Gross lesions alone are not sufficient** for a definitive diagnosis. Laboratory support is essential: correct samples in appropriate preservatives.
- A badly putrefied sample — or even a fresh sample in an unsuitable preservative — will not be useful.

“Doctors who perform necropsies, or who regularly witness postmortem examinations, learn to solve their doubts — unlike those not dealing with necropsy, whose answers are left floating in the air.”

— Dr. A. B. Shrivastav, 2014

“Carcass never tells a lie.”

— Dr. Satpati

Staff Required

Role	Number
Wildlife veterinarians	2–3
Supporting staff	4
Photographer	1
Sample collection specialist	1

3. Necropsy Objectives

1. Identify all possible **lesions of disease or abnormality**.
2. Correlate lesions with **clinical and laboratory findings**.
3. Understand the **sequence of disease events** (pathogenesis).
4. Provide information to **prevent or control the disease** in the healthy population.

4. Pre-Necropsy Evaluation

Before beginning, examine the **necropsy request letter** for:

- Specific lesions desired in particular organs or systems.
- Whether a **cosmetic** (for display) or **routine** necropsy is required.
- Special requests — sample collection for specific investigations, cultures, photographs, radiographs, etc.
- **Identify the carcass** positively before beginning — confirm the correct individual is being necropsied.
- Review all available **clinical and circumstantial history**.

PM examinations of Schedule I animals must be performed systematically, sincerely, and seriously — never as a routine chore.

5. Protocol for PM Examination — General Workflow

Step	Action	Notes
1	Letter from the competent authority	Legal authorisation before any PM is commenced
2	Facilities	Appropriate site — good daylight, space for large carcass, drainage
3	Measurements	Body length, girth, tusk length; use calibrated instruments
4	Observation — External then Internal	Systematic external examination before any incision; then internal organ systems
5	Gross interpretation	Record and interpret all gross lesions before collecting samples

Step	Action	Notes
6	Sample collection	Collect biological samples in correct preservatives immediately after gross review
7	Laboratory submission	Suitable samples dispatched with completed history form and transport certificate
8	Report writing	Tentative cause of death → confirmatory lab results → final determination

General Logistics — Non-Negotiable Requirements

Always perform in good daylight.

Use standard large-animal necropsy instruments (stainless steel, sharp).

Protective clothing (PPE) mandatory for all veterinary and support staff.

Maintain photographic and video documentation throughout.

6. Equipment Required

Documentation & Measurement Items

Item	Min. Qty
Copy of necropsy protocol	1
Hand-held GPS	1
Digital camera	1
Digital video camera	1
Clipboard	2
Measuring tape	1
Plastic ruler 15 cm (clear markings)	2
Vernier calipers	1
Portable weighing machine (250 kg)	1
Necropsy format (data sheets)	5
Ballpen	2
Pencil + rubber	1 set
Metal detector	1

Instruments (Stainless Steel — Best Quality)

Instrument	Min. Qty
Sharp high-quality necropsy knife (SS)	4
Skinning knife, curved (SS)	2
Autopsy knife, curved (SS)	2
Slicing knife, 22 mm blade (SS)	1
Knife sharpener stone or steel	1

Instrument	Min. Qty
Bard-Parker handle + blades (No. 20, 22, 26)	5 each
Small plain forceps	As needed
Artery forceps, straight 203 mm (SS)	2
Dissecting forceps, 200 mm (SS)	2
Large Mayo dissecting scissors, fine point 215 mm	2
Bone cutter, compound action 266 mm	1
Hack saw / bone saw blade 254 mm	1
Chisel 22 cm × 3 cm blade	1
Hammer, wrench end 200 mm	1
Portable autopsy saw (electric/battery)	1
Axe (roofing axe, SS) 60 mm blade	1
Spirit lamp (SS) or gas burner	1

7. External Examination

General Assessment

- **Body condition score** — assess fat cover, muscle mass.
- **Rigor mortis** — presence, degree, and stage.
- **External injuries** — characterize location, type, age of injury.
- **Documentation** — photograph every finding before disturbing.
- **Any other observation** relevant to circumstances of death.

Detailed Checklist (11 Points)

Check	What to Assess
1. Sex identification	Examine male/female external organs; note pregnancy, lactation, or dry status.
2. Nutritional status	Prominent ribs, depression in lumbar and buccal regions, distinct temporal fossa, and loose skin all indicate poor nutritional status.
3. Fractures & dislocations	Search for fractures and dislocations systematically where possible.
4. Postmortem changes	Assess algor mortis (cooling), rigor mortis (stiffening), livor mortis (gravitational blood pooling), and autolytic changes.
5. Mucous membranes	Examine for pallor (anaemia), congestion (hyperaemia), cyanosis (hypoxia), and inflammatory conditions.
6. Skin	Check for external parasites, cutaneous filariasis, pox lesions, oedematous swellings, emphysematous crepitation, injuries, burns, and warts.
7. Injuries	Look for bullet wounds, intraspecies fight injuries, large-carnivore mauling (especially in calves), and ankush (hook) injuries in captive elephants.
8. Dung bolus at rectum	Indicates digestive function, dentition status, and hydration — dry, hard pellets vs. loose, watery faeces are both informative.

Check	What to Assess
9. Natural orifices	Check all orifices for discharges. Examine oral cavity for FMD lesions, EEHV ulcers, anthrax signs, etc.
10. Temporal glands	Examine for musth secretions in adult males — relevant to circumstances of death (fighting, restraint-related stress).
11. Foot pads and nails	Check for injuries, cracks, foreign bodies, and overall condition.

8. Postmortem Changes

Change	Description	Timeline / Notes
Cadaveric lividity (Hypostatic congestion)	Irregular patches of dark discolouration in subcutaneous tissues on the dependent (lower) side — particularly prominent in large animals like elephants due to large blood volume. Must be distinguished from true haemorrhages.	Begins within 1–2 hours; fixed by 6–8 hours (cannot be moved by repositioning).
Rigor mortis	Progressive stiffening of muscle due to ATP depletion and actin-myosin crosslinking. Less marked in weak, emaciated, diseased, or septicemic animals.	Appears 1–4 hours post-death; lasts 16–24 hours; may persist up to 48 hours depending on temperature and nutritional state.
Autolytic changes	Cellular self-digestion by endogenous enzymes released from disrupted lysosomes. Rate depends on atmospheric moisture, temperature, and postmortem interval.	Begins immediately after death; visible as green discolouration, gas bloat, softening of tissues, and odour.

9. Dentition & Age Determination

Elephants develop **six successive sets of molars (M1–M6)** that erupt and wear sequentially throughout life. Each set replaces the previous as it wears down — unlike most mammals where molars erupt once. The number of **laminae** (enamel ridges) increases with each molar and can be used to estimate age.

Molar	Approx. Age at Eruption	Laminae (Enamel Ridges)
M1	~1 year	5
M2	~2 years	7
M3	~6 years	10
M4	~15 years	10
M5	~28 years	12
M6	~47 years	13

Tusk & Final Molar Note

Permanent tusks protrude beyond the lips at approximately 30 months and continue growing throughout life — tusk length can supplement age estimation. Once M6 wears out (typically 60+ years), the elephant can no longer chew and eventually dies of malnutrition.

10. Skinning the Carcass

In a healthy animal the skin is thick but flexible and moves easily over the underlying tissues. Proper skinning is essential before internal examination.

- Remove the skin of **one side** — from the entire lateral thoracic region down to the lower abdominal and other suspected regions.
- **Examine subcutaneous tissues and superficial muscles** for abscesses, wounds, haemorrhages, and other pathological conditions.
- Identify and examine all **superficial lymph nodes** before proceeding to the thoracic and abdominal cavities.

11. Internal Examination — Systematic Approach

System	Key Organs & Examination Points
Respiratory	Trachea — examine for mucus, haemorrhage, foreign body. Lungs — consistency, colour, pleural surface, cut surface (oedema, consolidation, haemorrhage).
Digestive	Pharynx, oesophagus, stomach, small intestine, large intestine, caecum, rectum, anus. Also: molar teeth and tongue, salivary glands, liver, pancreas. Note: the elephant is a simple-stomached (monogastric) animal. Stomach length ≈ 100–140 cm; diameter ≈ 40 cm. Examine stomach contents — food material, moisture, parasites, ulcers, gastritis.
Urogenital	Kidneys — size, colour, capsule, cut surface (cortex-medulla ratio). Bladder — distension, mucosa, content.
Reproductive	Uterus — size, content, endometrium. Ovaries / testes — as appropriate.
Musculoskeletal	Muscles — pallor, haemorrhage, necrosis, parasites. Bones — fractures, metabolic changes.
Nervous	Brain — meninges, congestion, herniation, grey/white matter changes.
Cardiovascular	Heart — size, epicardium, myocardium (cut surface), valves, great vessels.

Anatomical Note — Elephant Abdomen

Abdominal cavity approach: draw a vertical line from the last rib to the lower abdomen; make a vertical incision down to ground surface to open the cavity.
The elephant has TWENTY PAIRS of ribs (vs. 13 in most large mammals).

12. Lymph Node Examination

Superficial lymph nodes are particularly important for detecting infectious diseases such as **tuberculosis** and **EEHV**. Thick elephant skin makes external palpation difficult; systematic incision is required.

Superficial Lymph Nodes

Node	Location	Notes
Mandibular	At the ventral border of the lower jaw, just under the skin.	Incise between the mandibles and pull the tongue forward during oral cavity inspection.
Parotid	Just below the ear, near the parotid salivary gland.	Enlarged in head and neck infections.

Node	Location	Notes
Superficial cervical	On the lateral side of the neck, slightly cranial to the shoulder.	Drains superficial neck structures.
Prescapular	Just ahead of the shoulder joint.	Drains the forelimb and cranial thorax.
Femoral	Close to the head of the femur.	Drains the hindlimb.

Deep / Internal Lymph Nodes

Node	Location	Clinical Significance
Bronchial (Tracheobronchial)	At the carina — where the bronchi bifurcate from the trachea.	MUST be sampled for mycobacterial culture (TB). First site of lung lymph drainage; enlarged and caseous in TB.
Hepatic & splenic	Near the liver (hepatic hilus) and spleen respectively.	Checked for systemic infection, haemorrhage, and reactive hyperplasia.
Mesenteric	Clusters distributed in the mesentery throughout the bowel.	Filter lymph from the intestines; trap bacteria, viruses, and antigens before they enter the bloodstream. Key in enteric infections.

13. Histopathology Highlight — EEHV

Elephant Endotheliotropic Herpesvirus (EEHV) is one of the most dangerous pathogens of young Asian elephants. Histopathological examination of affected endothelial tissues — particularly from the tongue, heart, and trunk tip — may reveal **intranuclear inclusion bodies**, the hallmark of herpesvirus infection.

EEHV — Histopathology & Diagnosis

Intranuclear inclusion bodies in vascular endothelium = EEHV until proven otherwise.

Collect trunk wash, oral and rectal swabs, and fresh endothelial-rich tissue (tongue tip, trunk tip) for PCR confirmation.

A T-cell-inducing heterologous vaccine against EEHV is currently under trial study.

14. Differential Diagnosis — Sudden Onset, Rapid Death

Disease	Characteristic Finding	Key Samples
Haemorrhagic septicaemia (Pasteurellosis)	Extensive haemorrhagic areas throughout the carcass; acute gelatinous oedema in subcutaneous tissues of the neck and brisket.	Swabs, blood smear, culture from lymph node and lung.
Salmonellosis	Watery diarrhoea with or without blood; small intestinal mucosal haemorrhage and necrosis; mesenteric lymph node enlargement.	Intestinal loop culture; fresh faeces; mesenteric LN.
Enterotoxaemia (Clostridial)	Profuse diarrhoea leading to rapid dehydration and death; bloat; haemorrhagic intestinal contents.	Intestinal loop in formalin; fresh intestinal content for toxin assay.
Encephalomyocarditis (EMCV)	Cardiac muscle pallor and necrosis; respiratory signs may be observed; sudden death in young/stressed elephants.	Heart tissue (fresh and fixed); serum for virus isolation/PCR.

15. Case Example — Rabies in Semi-Captive Elephant (Panna TR)

A semi-captive elephant at Panna Tiger Reserve presented with the following sequence of clinical signs:

Time Point	Clinical Event
Initially	Off feed, restlessness, excessive drooling of saliva.
Day 3	Restricted jaw movement, protrusion of tongue.
Day 4	Hyperaemic conjunctiva (intense reddening of the eyes).
Day 5	Hydrophobia; loud but markedly changed/abnormal trumpeting.
Day 6	Lateral recumbency and death.
Post-death	Proper carcass disposal under biosafety protocols.

Rabies — Sample & Biosafety Notes

Rabies in elephants is rare but documented. Submit fresh brain tissue in two portions: (1) half in 10% NBF for histopathology (Negri body detection), (2) half fresh/frozen for fluorescent antibody test (FAT) or RT-PCR. Handle with full Level-3 PPE. Carcass disposal must follow zoonotic disease biosafety protocols.

16. Other Conditions of Note

Condition	Notes for PM Examination
Tuberculosis (TB)	Gross lesions — caseous nodules in lung and tracheobronchial lymph nodes — documented in free-ranging Asian elephants. Always sample bronchial lymph nodes for mycobacterial culture.
HCN (hydrocyanic acid) poisoning	Common in herbivores fed sorghum/jowar species; cyanogenic glycoside content highest in young, rapidly growing leaves and stalks (after autumn rains or crop regrowth). Managed with sodium thiosulphate + supportive therapy. (Ref: Shrivastav, Zoo Print, Vol. XXV, June 2010.)
Decomposed / waterlogged carcass	Severely autolysed or submerged carcasses may make assessment "difficult to advise" — tissue architecture is destroyed, cultures contaminated, and toxicological results unreliable. Emphasises the value of early detection.

Key Takeaways

Chapter Summary — Postmortem Examination of the Asian Elephant

1. A wildlife necropsy is only as good as its protocol, facilities, and operator experience. Always work in good daylight with complete PPE.
2. Gross findings alone rarely yield a definitive diagnosis — confirmatory diagnosis depends on correctly collected and preserved laboratory samples.
3. For elephants: use molars/tusks for age estimation; remember the monogastric stomach (100–140 cm); and always note the 20 pairs of ribs.
4. Prioritise tracheobronchial lymph node sampling for TB and trunk wash / endothelial tissue for EEHV PCR confirmation.
5. Cadaveric lividity must be distinguished from true haemorrhagic lesions.
6. Rabies carries zoonotic risk — handle with Level-3 PPE; dispose of carcass under biosafety protocols.
7. "The carcass never tells a lie" — but only if the necropsy is performed promptly, systematically, and with full documentation.